PARISHRAM 2.0

CLASS 12TH 2026

Inverse Trigonometric Functions

MATHS | LECTURE-5

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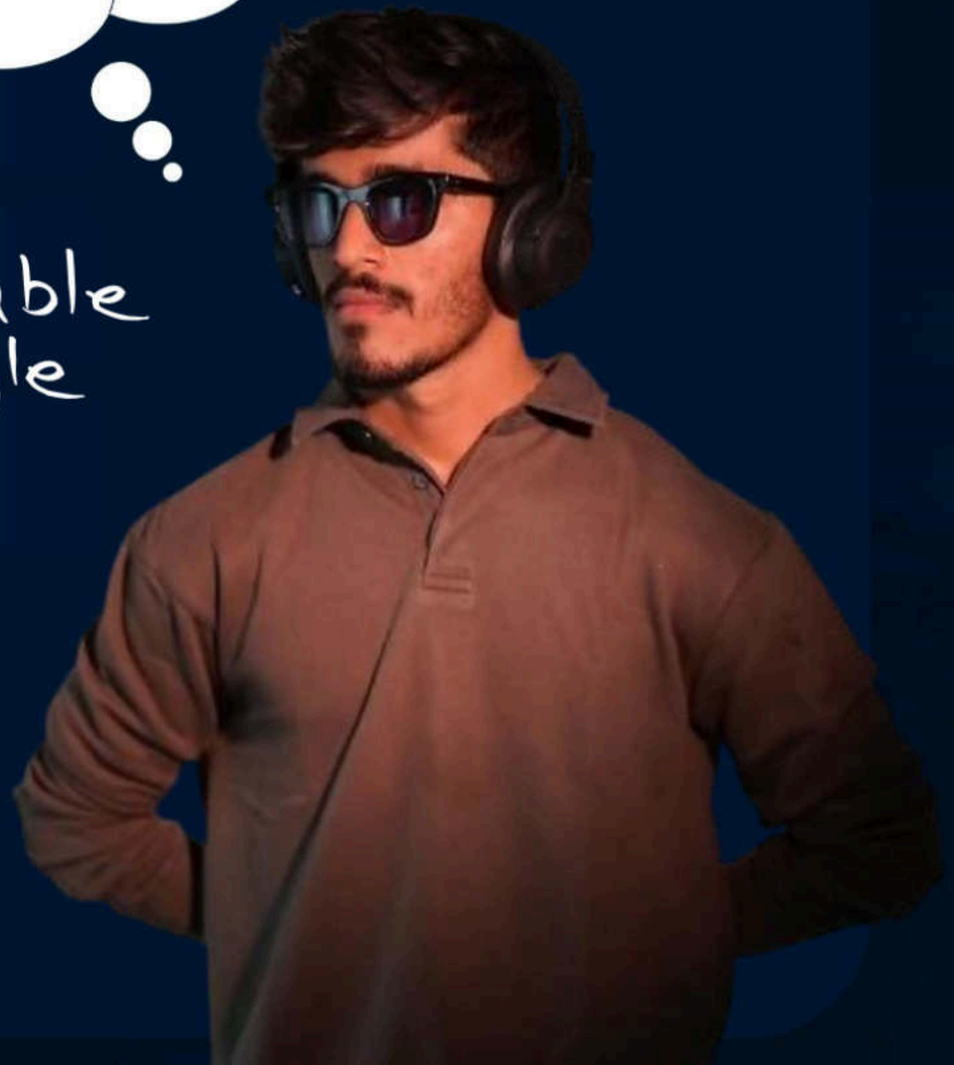
TOPICS TO BE COVERED

1. Substitution → ITF
2. Triple Angled Formulas
3. Questions on it

sin2θ — Double
cos2θ — angle

→ sin3θ
→ cos3θ
→ tan3θ

Why Fear
When Rolex Sir
is Here





Inverse Trigo

- 11th ke Trigo ✓
- 12th (Property Bunch)

} Yaad hogaya ✓

Questions ✓

Class 11th Trigo Formula for 12th

- $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$
- $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$
- $\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$



Inverse Trigo

Substitution

Expression	Substitution
$a^2 + x^2$	$x = a\tan\theta$ OR $x = a\cot\theta$
$a^2 - x^2$	$x = a\sin\theta$ OR $x = a\cos\theta$
$x^2 - a^2$	$x = a\sec\theta$ OR $x = a\operatorname{cosec}\theta$
$1 + x^2$	$x = \tan\theta$ $x = \cot\theta$
$1 - x^2$	$x = \sin\theta$ $x = \cos\theta$
$x^2 - 1$	$x = \sec\theta$ $x = \operatorname{cosec}\theta$

a = 1

Farak kya h ??



QUESTION

$$\cot^{-1}\left\{\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right\}$$

$\sin 2\theta = 2\sin\theta\cos\theta$

$\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$



$1+\sin x = \sin^2\frac{x}{2} + \cos^2\frac{x}{2} + 2\sin\frac{x}{2}\cos\frac{x}{2}$

$a^2 + b^2 + 2ab = (a+b)^2$

$1+\sin x = \left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)^2$

$1-\sin x = \cos^2\frac{x}{2} + \sin^2\frac{x}{2} - 2\sin\frac{x}{2}\cos\frac{x}{2}$

$a^2 + b^2 - 2ab$

$1-\sin x = \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2$

$$\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right)$$

$$1+\sin x = \left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2$$

$$1-\sin x = \left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2$$

$$\cot^{-1}\left(\frac{\sqrt{\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2}+\sqrt{\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2}}{\sqrt{\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2}-\sqrt{\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2}}\right)$$

$$\cot^{-1}\left(\frac{\sin\frac{x}{2}+\cos\frac{x}{2}+\cos\frac{x}{2}-\sin\frac{x}{2}}{\sin\frac{x}{2}+\cos\frac{x}{2}-\cos\frac{x}{2}+\sin\frac{x}{2}}\right)$$

$$\cot^{-1}\left(\frac{2\cos\frac{x}{2}}{2\sin\frac{x}{2}}\right) = \cot^{-1}\left(\frac{\cos\frac{x}{2}}{\sin\frac{x}{2}}\right)$$

$$\cot^{-1}\left(\cot\frac{x}{2}\right) = \boxed{\frac{x}{2}}$$

QUESTION (Kal vala type)

Find the value of $\sin\left(\cos^{-1}\frac{4}{5}+\tan^{-1}\frac{2}{3}\right)$

$$\sin\left(\underbrace{\cos^{-1}\frac{4}{5}}_{\alpha}+\underbrace{\tan^{-1}\frac{2}{3}}_{\beta}\right)$$

$$= \sin(\alpha+\beta)$$

$$= \sin\alpha\cos\beta+\cos\alpha\sin\beta$$

$$= \frac{3}{5}\times\frac{3}{\sqrt{13}}+\frac{4}{5}\times\frac{2}{\sqrt{13}}$$

$$= \frac{9}{5\sqrt{13}}+\frac{8}{5\sqrt{13}}=\frac{17}{5\sqrt{13}}$$

$\beta=\tan^{-1}\left(\frac{2}{3}\right)$

$\tan\beta=\frac{2}{3}$

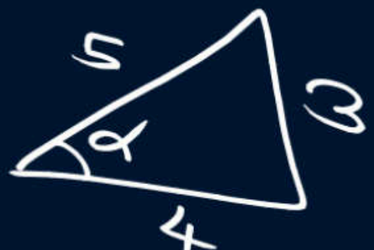


$\sin\beta=\frac{2}{\sqrt{13}}$

$\cos\beta=\frac{3}{\sqrt{13}}$

$\alpha=\cos^{-1}\left(\frac{4}{5}\right)$

$\cos\alpha=\frac{4}{5}$



$\sin\alpha=\frac{3}{5}$



QUESTION

$x = \tan\theta$

$\tan^{-1}x = \theta$

$\frac{\theta}{2} = \frac{\tan^{-1}x}{2}$

Write the following functions in the simplest form:

$$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right), x \neq 0$$



$x = \tan\theta$

$$\tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$$

$$\tan^{-1}\left(\frac{\sqrt{\sec^2\theta}-1}{\tan\theta}\right)$$

$$\tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right)$$

$$\tan^{-1}\left(\frac{\frac{1}{\cos\theta}-1}{\frac{\sin\theta}{\cos\theta}}\right)$$

$$\tan^{-1}\left(\frac{1-\cos\theta}{\frac{\sin\theta}{\cos\theta}}\right)$$

$$\tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

$$= \tan^{-1}\left(\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right)$$

$$= \tan^{-1}\left(\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\right)$$

$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$= \frac{\theta}{2}$$

$x = a\sin\theta$

$\frac{x}{a} = \sin\theta \longrightarrow \sin^{-1}\left(\frac{x}{a}\right) = \theta$

QUESTION

Write the following functions in the simplest form:

$$\tan^{-1}\left\{\frac{x}{\sqrt{a^2 - x^2}}\right\}, |x| < a$$



$x = a\sin\theta$

$$\tan^{-1}\left(\frac{a\sin\theta}{\sqrt{a^2-(a\sin\theta)^2}}\right) = \tan^{-1}\left(\frac{a\sin\theta}{a\sqrt{1-\sin^2\theta}}\right)$$

$$\tan^{-1}\left(\frac{a\sin\theta}{\sqrt{a^2-a^2\sin^2\theta}}\right) = \tan^{-1}\left(\frac{a\sin\theta}{a\sqrt{\cos^2\theta}}\right)$$

$$\tan^{-1}\left(\frac{a\sin\theta}{\sqrt{a^2(1-\sin^2\theta)}}\right) = \tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right) = \tan^{-1}(\tan\theta)$$

$$= \theta = \sin^{-1}\left(\frac{x}{a}\right)$$

$\frac{x}{a} = \tan\theta \quad \tan^{-1}\left(\frac{x}{a}\right) = \theta$

QUESTION

Write the following functions in the simplest form:

$$\tan^{-1}\left(\frac{3a^2x - x^3}{a^3 - 3ax^2}\right), a > 0; \frac{-a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$$



$x = a\tan\theta$

$$\tan^{-1}\left(\frac{3a^2(a\tan\theta) - (a\tan\theta)^3}{a^3 - 3a(a\tan\theta)^2}\right)$$

$$\tan^{-1}\left(\frac{3a^3\tan\theta - a^3\tan^3\theta}{a^3 - 3a^3\tan^2\theta}\right)$$

$$\tan^{-1}\left(\frac{a^3(3\tan\theta - \tan^3\theta)}{a^3(1 - 3\tan^2\theta)}\right)$$

$$= \tan^{-1}\left(\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}\right)$$

$$= \tan^{-1}(\tan 3\theta)$$

$$= 3\theta = 3\tan^{-1}\frac{x}{a}$$

Q) $\tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$

$\frac{1}{2}\cos^{-1}x = \theta$

$\boxed{x = \cos 2\theta}$ $\quad \cos^{-1}x = 2\theta$

$\tan^{-1}\left(\frac{\sqrt{1+\cos 2\theta}-\sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta}+\sqrt{1-\cos 2\theta}}\right)$

$\tan^{-1}\left(\frac{\sqrt{2\cos^2\theta}-\sqrt{2\sin^2\theta}}{\sqrt{2\cos^2\theta}+\sqrt{2\sin^2\theta}}\right)$

$\tan^{-1}\left(\frac{\sqrt{2}\cos\theta-\sqrt{2}\sin\theta}{\sqrt{2}\cos\theta+\sqrt{2}\sin\theta}\right)$

$\tan^{-1}\left(\frac{\sqrt{2}(\cos\theta-\sin\theta)}{\sqrt{2}(\cos\theta+\sin\theta)}\right)$

$\tan^{-1}\left(\frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}\right)$

$\tan^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right)$

$\tan^{-1}\left(\frac{\tan\frac{\pi}{4}-\tan\theta}{1+\tan\frac{\pi}{4}\tan\theta}\right)$

$= \tan^{-1}\left(\tan\left(\frac{\pi}{4}-\theta\right)\right) = \frac{\pi}{4}-\theta$

THANK YOU